Machine M-17 Temperature Disagreement Investigation

Question: Two inspection reports record different temperatures for machine M-17 at the same time (92°C vs 76°C). Can the disagreement be explained, and does the later sensor recalibration account for it?

Generated: 2026-10-10T22:14:24.335851+00:00 Confidence: 0.43

Investigation id: 1 Status: completed

Demonstration data for evaluation purposes.

Investigation Title and Identifier

Machine M-17 Temperature Disagreement Investigation (id 1).

Investigation Title

Machine M-17 Temperature Disagreement Investigation

Objective

Two inspection reports record different temperatures for machine M-17 at the same time (92°C vs 76°C). Can the disagreement be explained, and does the later sensor recalibration account for it?

Documents Analyzed

3 document(s) were analyzed. These are demonstration documents.

? [Inspection_Report_A.txt](/documents/1) (txt, 115 bytes, 1 claim(s), status=processed)

? [Inspection_Report_B.txt](/documents/2) (txt, 115 bytes, 1 claim(s), status=processed)

? [Maintenance_Log_C.txt](/documents/3) (txt, 91 bytes, 1 claim(s), status=processed)

Claims Extracted

3 claim(s) were extracted and are treated as reported observations, not established facts.

? [Inspection_Report_A.txt](/documents/1) claim 1: "[DEMONSTRATION DATA] Inspection Report A Machine M-17 reached a temperature of 92°C during inspection at 10:00 AM." (numeric=92.0 °C) [source: Inspection_Report_A.txt]

? [Inspection_Report_B.txt](/documents/2) claim 2: "[DEMONSTRATION DATA] Inspection Report B Machine M-17 reached a temperature of 76°C during inspection at 10:00 AM." (numeric=76.0 °C) [source: Inspection_Report_B.txt]

? [Maintenance_Log_C.txt](/documents/3) claim 3: "[DEMONSTRATION DATA] Maintenance Log C The temperature sensor was recalibrated at 10:30 AM." [source: Maintenance_Log_C.txt]

Key Observations

? Inspection_Report_A.txt: [DEMONSTRATION DATA] Inspection Report A Machine M-17 reached a temperature of 92°C during inspection at 10:00 AM. (source: Inspection_Report_A.txt)

? Inspection_Report_B.txt: [DEMONSTRATION DATA] Inspection Report B Machine M-17 reached a temperature of 76°C during inspection at 10:00 AM. (source: Inspection_Report_B.txt)

? Maintenance_Log_C.txt: [DEMONSTRATION DATA] Maintenance Log C The temperature sensor was recalibrated at 10:30 AM. (source: Maintenance_Log_C.txt)

Conflicting Claims

1 potential conflict(s) were identified.

? Conflict 1: 'temperature' is reported as 92°C and 76°C within the same time period with no distinguishing context (17% difference).

Conflict Classifications

? Conflict 1: type=numerical_disagreement, severity=medium, status=human_review, claims 1 vs 2

Evidence Supporting Each Relevant Claim

Evidence explicitly recorded against each claim that takes part in a conflict.

? Claim 1: evidence 1 [measurement] Claim A states: "[DEMONSTRATION DATA] Inspection Report A Machine M-17 reached a temperature of 92°C during inspection at 10:00 AM." (source:

Inspection_Report_A.txt)
? Claim 2: evidence 2 [measurement] Claim B states: "[DEMONSTRATION DATA] Inspection Report B Machine M-17 reached a temperature of 76°C during inspection at 10:00 AM." (source: Inspection_Report_B.txt)

Evidence Contradicting Each Relevant Claim

Evidence recorded against the opposing claim in each recorded conflict. No contradiction is inferred beyond what the conflict identifies.

? Claim 1: evidence 2 [measurement] Claim B states: "[DEMONSTRATION DATA] Inspection Report B Machine M-17 reached a temperature of 76°C during inspection at 10:00 AM." (Recorded conflict 1 places this claim against claim 2.) (source: Inspection_Report_B.txt)
? Claim 2: evidence 1 [measurement] Claim A states: "[DEMONSTRATION DATA] Inspection Report A Machine M-17 reached a temperature of 92°C during inspection at 10:00 AM." (Recorded conflict 1 places this claim against claim 1.) (source: Inspection_Report_A.txt)

Competing Hypotheses

4 explanation(s) were proposed. These are inferred explanations, not observed facts.

? Conflict 1: "Sensor or measurement-source issue" (status=review_required, support=0.72, origin=inferred)
? Conflict 1: "Measurement or transcription error" (status=inconclusive, support=0.50, origin=inferred)
? Conflict 1: "Genuine difference under otherwise similar conditions" (status=inconclusive, support=0.40, origin=inferred)
? Conflict 1: "Incomplete reporting / missing context" (status=inconclusive, support=0.50, origin=inferred)

Hypothesis Ranking

Competing hypotheses are ranked by documented evidence weight. Scores are bounded support indicators, not calibrated probabilities.

? Conflict 1 rank 1: "Sensor or measurement-source issue" (support=0.72, confidence=0.68, +weight=0.81, -weight=0.00, distinct sources=1, direct evidence=True, margin to next=0.22)
? Conflict 1 rank 2: "Measurement or transcription error" (support=0.50, confidence=0.40, +weight=0.00, -weight=0.00, distinct sources=0, direct evidence=False, margin to next=0.00)
? Conflict 1 rank 3: "Incomplete reporting / missing context" (support=0.50, confidence=0.40, +weight=0.00, -weight=0.00, distinct sources=0, direct evidence=False, margin to next=0.10)
? Conflict 1 rank 4: "Genuine difference under otherwise similar conditions" (support=0.40, confidence=0.32, +weight=0.00, -weight=0.32, distinct sources=0, direct evidence=False)

Evidence Supporting Each Hypothesis

? Evidence 3 supports hypothesis 1: Document 'Maintenance_Log_C.txt' contains a calibration record: "[DEMONSTRATION DATA] Maintenance Log C The temperature sensor was recalibrated at 10:30 AM."

Evidence Contradicting Each Hypothesis

? Evidence 4 contradicts hypothesis 3: No operating-condition data was found in the available documents for this conflict.

Verification Results

Reliability/relevance are bounded quality indicators; 'unverified' means no verification step has been recorded.

? Evidence 1 [measurement] verification=verified (reliability=0.70, relevance=0.60, weight=0.70)
? Evidence 2 [measurement] verification=verified (reliability=0.70, relevance=0.60, weight=0.70)
? Evidence 3 [calibration_record] verification=verified (reliability=0.80, relevance=0.80, weight=0.81)
? Evidence 4 [missing] verification=unsupported (reliability=0.20, relevance=0.50, weight=0.32)

Final Conclusion or Inconclusive Status

Each conclusion is the system's adjudication for a recorded conflict. No definitive conclusion was produced; the investigation is inconclusive.

? Conflict 1 -> human_review_required (confidence 0.43): The evidence about 'numerical_disagreement' could not be resolved automatically from the available documents. Human review is recommended. Best-supported explanation: "Sensor or measurement-source issue" (support 0.72). The two readings may come from different measurement sources, or the sensor may have been miscalibrated or drifting, so the disagreement reflects instrumentation rather than a real change in the machine. Supporting evidence: Evidence 3 (calibration_record from Maintenance_Log_C.txt): Document 'Maintenance_Log_C.txt' contains a calibration record: "[DEMONSTRATION DATA] Maintenance Log C The temperature sensor was recalibrated at 10:30 AM.".

Confidence and its Limitations

Overall confidence (mean of conclusions): 0.43. Confidence is the mean of the per-conflict conclusion support scores. It is a bounded support indicator derived from documented evidence weights, NOT a statistically calibrated probability. A higher value means more (reliable, direct, source-diverse) supporting evidence relative to contradicting evidence; it does not mean 'X% likely to be true'.

? Confidence scores are bounded support indicators, not calibrated probabilities.
? Only information present in the uploaded documents was considered.

Unresolved Questions

? Calibration history of the sensor and the measurement source for each reading.
? Assumption: A sensor or instrument was used and may not be fully trustworthy.
? Independent confirmation of either value.
? Assumption: Only one of the values is correct.
? Timing, location and conditions of each measurement.
? Assumption: The reports omit relevant details.
? Conditions and exact timing for each measurement.
? Assumption: Conditions were comparable.

Evidence Gaps and Uncertainty

Missing evidence that limits confidence in any conclusion.

? Conflict 1: No operating-condition data was found in the available documents for this conflict.
? Conflict 1: Calibration history of the sensor and the measurement source for each reading.
? Conflict 1: Independent confirmation of either value.
? Conflict 1: Conditions and exact timing for each measurement.
? Conflict 1: Timing, location and conditions of each measurement.

Human Review Status

Status: awaiting review (0 decision(s) recorded).

? Conflict 1 is awaiting human review.

Recommended Next Steps

? Route conflict 1 to a human reviewer with the attached evidence.

Human Review Notes

This report was produced by an automated evidence-driven system. It supports, but does not replace, professional judgement.

? Conflict 1 requires human review.

Human Review Decisions

Recorded human decisions on this investigation's conclusions.

? No human review decisions have been recorded yet.

Agent Activity Summary

7 agent action(s) were recorded.

? Evidence Collector: 1 action(s), 1 success, 0 failed, 12 ms total
? Conflict Detector: 2 action(s), 1 success, 0 failed, 5 ms total
? Hypothesis Generator: 1 action(s), 1 success, 0 failed, 10 ms total
? Investigator: 1 action(s), 1 success, 0 failed, 6 ms total
? Verification Agent: 1 action(s), 1 success, 0 failed, 7 ms total
? Decision Agent: 1 action(s), 1 success, 0 failed, 4 ms total

Limitations and Timestamps

All times are UTC. Confidence is a bounded support indicator, not a probability.

? Created: 2026-10-10T22:14:24.030732+00:00
? Started: 2026-10-10T22:14:24.035023+00:00
? Completed: 2026-10-10T22:14:24.313814+00:00
? Report generated: 2026-10-10T22:14:24.335851+00:00
? Duration: 0.3 second(s)
? Limitation: Confidence scores are bounded support indicators, not calibrated probabilities.
? Limitation: Only information present in the uploaded documents was considered.

Investigation Stopping Reason

Stopped because no eligible agent could provide further useful information.

Audit Trail

Every recorded blackboard change for this investigation, newest last.

? #1 2026-10-10T22:14:24.057810+00:00 [claim None] extracted by agent: Extracted 1 rule-based and 0 LLM-suggested claim(s) and 1 observation(s) from 'Inspection_Report_A.txt'.
? #2 2026-10-10T22:14:24.060315+00:00 [claim None] extracted by agent: Extracted 1 rule-based and 0 LLM-suggested claim(s) and 1 observation(s) from 'Inspection_Report_B.txt'.
? #3 2026-10-10T22:14:24.061884+00:00 [claim None] extracted by agent: Extracted 1 rule-based and 0 LLM-suggested claim(s) and 1 observation(s) from 'Maintenance_Log_C.txt'.
? #4 2026-10-10T22:14:24.124905+00:00 [conflict 1] created by agent: Detected numerical_disagreement (severity=medium): 'temperature' is reported as 92°C and 76°C within the same time period with no distinguishing context (17% difference).
? #5 2026-10-10T22:14:24.160308+00:00 [evidence 1] created by agent: [measurement] Claim A states: "[DEMONSTRATION DATA] Inspection Report A Machine M-17 reached a temperature of 92°C during inspection at 10:00 AM."
? #6 2026-10-10T22:14:24.161410+00:00 [evidence 2] created by agent: [measurement] Claim B states: "[DEMONSTRATION DATA] Inspection Report B Machine M-17 reached a temperature of 76°C during inspection at 10:00 AM."
? #7 2026-10-10T22:14:24.162370+00:00 [hypothesis 1] created by agent: Proposed hypothesis 'Sensor or measurement-source issue' for conflict 1.
? #8 2026-10-10T22:14:24.162795+00:00 [hypothesis 2] created by agent: Proposed hypothesis 'Measurement or transcription error' for conflict 1.
? #9 2026-10-10T22:14:24.163166+00:00 [hypothesis 3] created by agent: Proposed hypothesis 'Genuine difference under otherwise similar conditions' for conflict 1.
? #10 2026-10-10T22:14:24.163491+00:00 [hypothesis 4] created by agent: Proposed hypothesis 'Incomplete reporting / missing context' for conflict 1.
? #11 2026-10-10T22:14:24.204703+00:00 [evidence 3] created by agent: [calibration_record] Document 'Maintenance_Log_C.txt' contains a calibration record: "[DEMONSTRATION DATA] Maintenance Log C The temperature sensor was recalibrated at 10:30 AM."
? #12 2026-10-10T22:14:24.205981+00:00 [evidence 4] created by agent: [missing] No operating-condition data was found in the available documents for this conflict.
? #13 2026-10-10T22:14:24.230676+00:00 [hypothesis 1] verified by agent: Hypothesis 'Sensor or measurement-source issue' evaluated as review_required (support=0.72, confidence=0.68).

? #14 2026-10-10T22:14:24.231752+00:00 [hypothesis 2] verified by agent: Hypothesis 'Measurement or transcription error' evaluated as inconclusive (support=0.50, confidence=0.40).
? #15 2026-10-10T22:14:24.233050+00:00 [hypothesis 3] verified by agent: Hypothesis 'Genuine difference under otherwise similar conditions' evaluated as inconclusive (support=0.40, confidence=0.32).
? #16 2026-10-10T22:14:24.233854+00:00 [hypothesis 4] verified by agent: Hypothesis 'Incomplete reporting / missing context' evaluated as inconclusive (support=0.50, confidence=0.40).
? #17 2026-10-10T22:14:24.234594+00:00 [evidence None] verification_completed by agent: Verified 4 evidence item(s) for conflict 1.
? #18 2026-10-10T22:14:24.274164+00:00 [conclusion 1] created by agent: Conclusion for conflict 1: human_review_required (confidence 0.43).

Report Warnings

Data-completeness notices that affect how this report should be read.

? At least one conclusion is inconclusive or requires human review; the report does not assert a definitive finding.
? A conflict is awaiting human review; no decision has been recorded yet.